

Internship Report - Task 1

1. Introduction

This report documents the work completed for Task 1 of the Redynox Cyber Security Internship: Introduction to Network Security Basics. The objective was to understand common network threats, implement basic security measures on a small network, monitor network traffic using Wireshark, and document findings.

2. Task Overview

2.1 Task 1: Introduction to Network Security Basics

Objective of the task:

- Understand different network threats (viruses, worms, trojans, phishing) and basic security concepts (firewalls, encryption, secure configurations).

Tools and methods used:

- Tools: Windows Defender Firewall, Router Web Interface, Nmap, Wireshark, Windows OS.
- Methods: Router hardening, firewall enablement, port scanning, packet capture and analysis.

Steps taken to complete the task:

1. Reviewed network security concepts: viruses, worms, trojans, phishing; firewalls; encryption; secure configs.
2. Logged into router web interface (usually 192.168.1.1) and changed default admin password.
3. Configured Wi-Fi to use WPA2 with AES on both 2.4GHz and 5GHz bands and disabled WPS.
4. Enabled Windows Defender Firewall on the host VM for host-level protection.
5. Performed network reconnaissance using Nmap to check open/filtered ports and services.
6. Installed Wireshark and captured network traffic on the active interface (DNS, HTTP, TLS).
7. Filtered and analyzed packets (dns, http, tls), inspected Client Hello and Server Hello messages, and saved captures.
8. Saved all evidence (screenshots and pcapng) and prepared this report.

Challenges faced and how they were overcome:

- Challenge: Capturing Wi-Fi monitor-mode traffic can be limited on Windows VMs due to adapter restrictions.

Resolution: Captured traffic on the host/bridged adapter and inside the VM where necessary. For TLS analysis, SNI in Client Hello was used to identify domains without needing to decrypt traffic.

- Challenge: Ensuring the captured packets included both unencrypted and encrypted traffic.

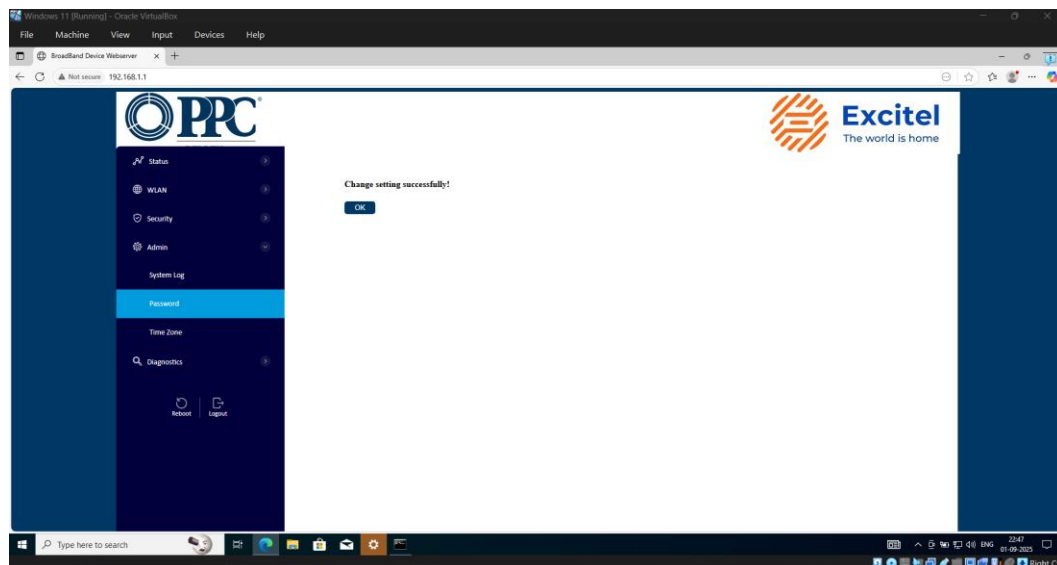
Resolution: Used a combination of visiting plain HTTP (neverssl.com) for HTTP captures and standard HTTPS sites for TLS handshakes.

Results and outcomes:

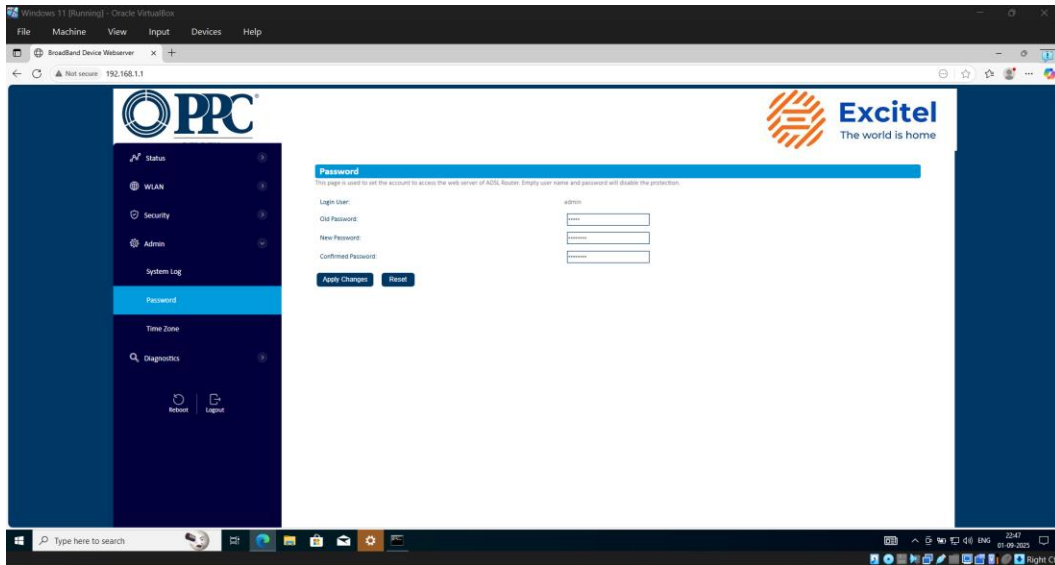
- Router configuration: default admin password changed, WPA2/AES enabled, WPS disabled (reduces attack surface).
- Firewall: Windows Defender Firewall enabled for host-level blocking of unauthorized inbound traffic.
- Nmap: Scans showed ports mostly filtered/closed, indicating limited exposure of services.
- Wireshark: Successful captures for DNS, HTTP (GET/200 OK), and TLS (Client Hello with SNI and Server Hello with selected cipher suite). This validated network monitoring capability and confirmed encrypted traffic for HTTPS.

Evidence & Screenshots

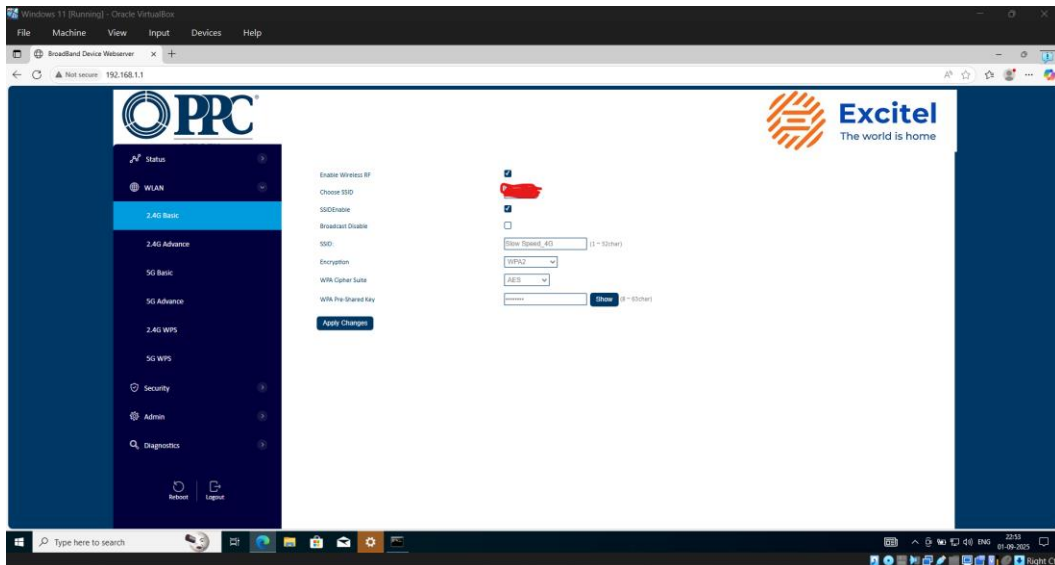
Screenshot 2025-09-01 224728.png



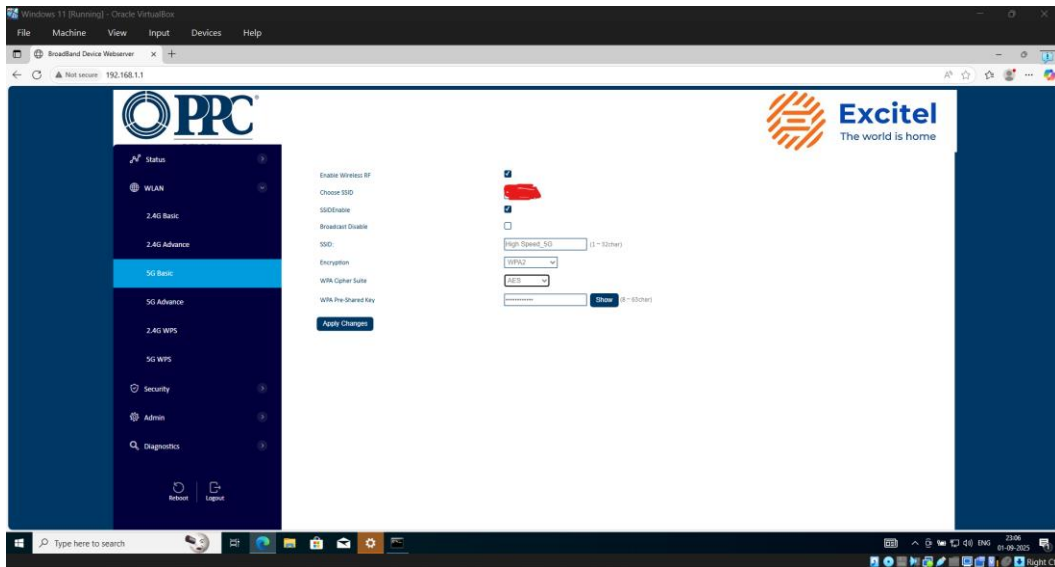
Screenshot 2025-09-01 224807.png



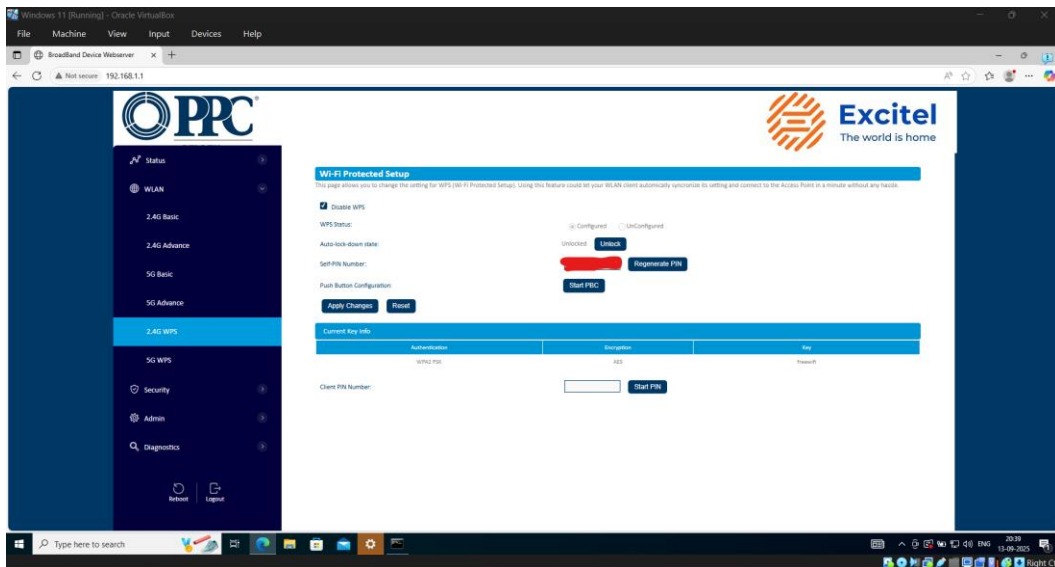
Screenshot 2025-09-01 225438.png



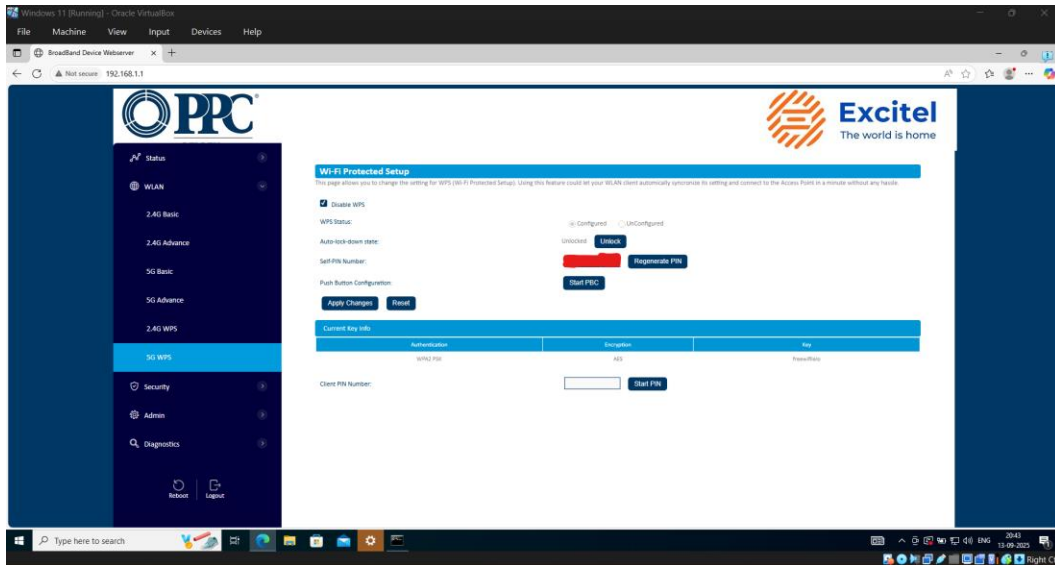
Screenshot 2025-09-01 230638.png



Screenshot 2025-09-13 204027.png



Screenshot 2025-09-13 204341.png



Screenshot 2025-09-13 224121.png

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Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

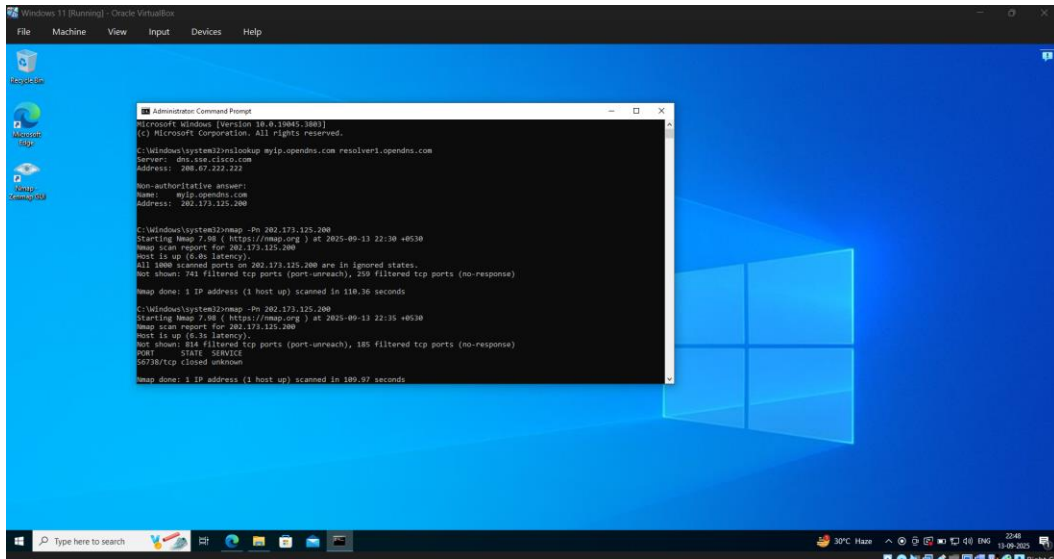
C:\Windows\system32>nslookup myip.opendns.com resolver1.opendns.com
Server: dns.sse.cisco.com
Address: 208.67.222.222

Non-authoritative answer:
Name: myip.opendns.com
Address: 202.173.125.200

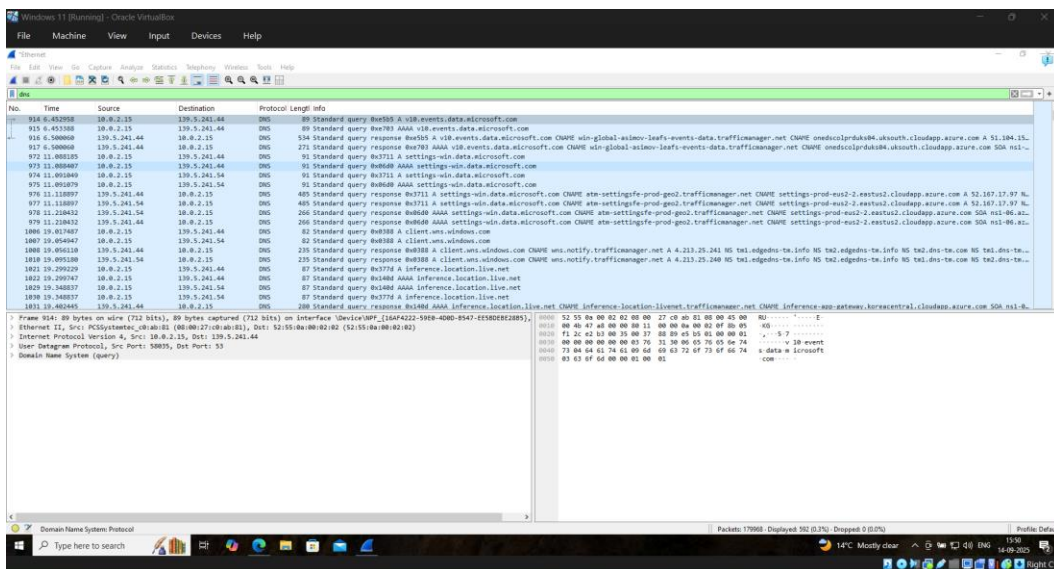
C:\Windows\system32>nmap -Pn 202.173.125.200
Starting Nmap 7.98 ( https://nmap.org ) at 2025-09-13 22:30 +0530
Nmap scan report for 202.173.125.200
Host is up (6.0s latency).
All 1000 scanned ports on 202.173.125.200 are in ignored states.
Not shown: 741 filtered tcp ports (port-unreach), 259 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 110.36 seconds

C:\Windows\system32>nmap -Pn 202.173.125.200
Starting Nmap 7.98 ( https://nmap.org ) at 2025-09-13 22:35 +0530
Nmap scan report for 202.173.125.200
Host is up (6.3s latency).
Not shown: 814 filtered tcp ports (port-unreach), 185 filtered tcp ports (no-response)
PORT      STATE SERVICE
56738/tcp closed unknown
Nmap done: 1 IP address (1 host up) scanned in 109.97 seconds
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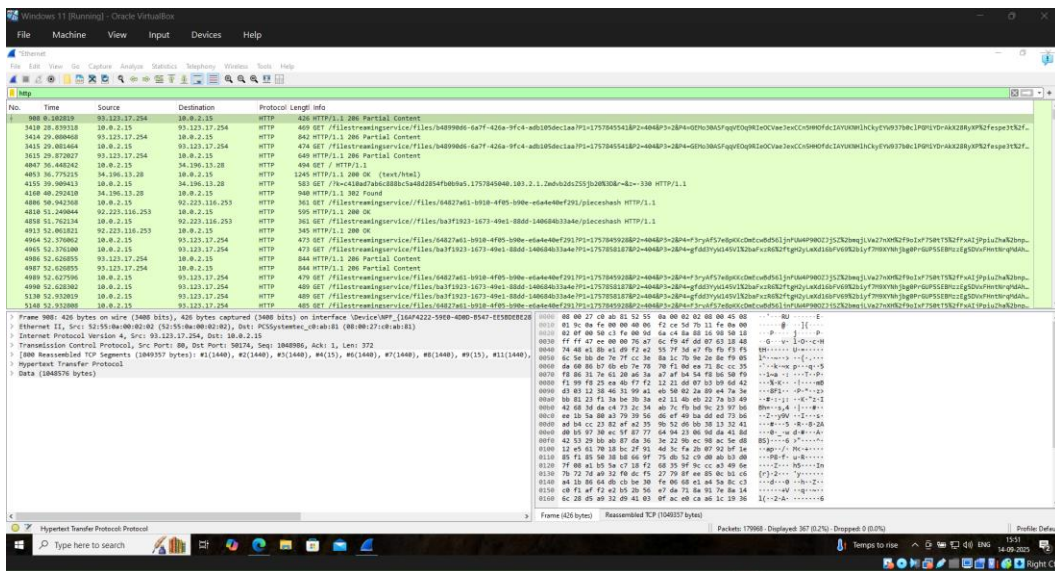
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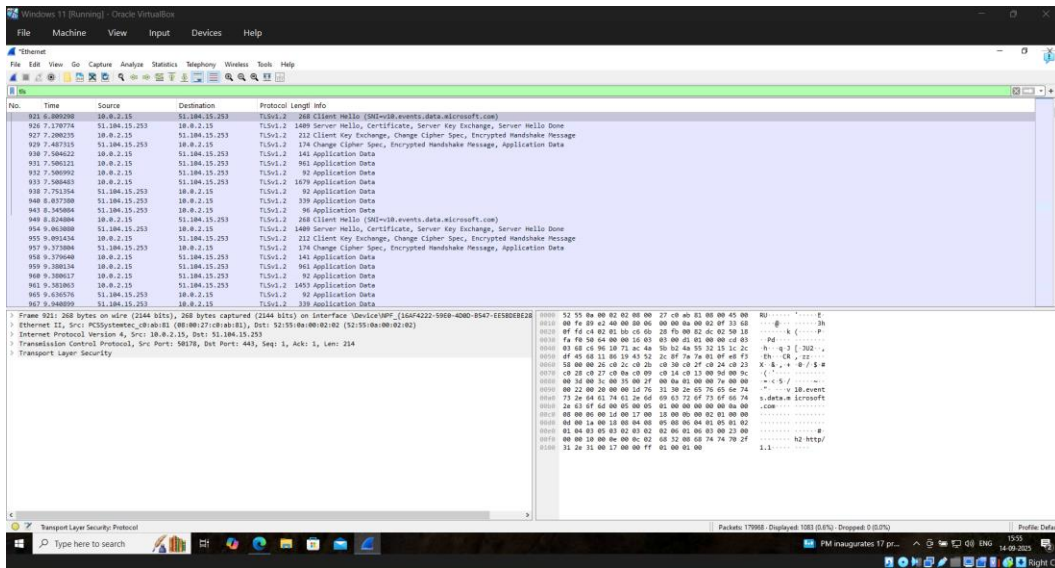
Screenshot 2025-09-14 155019.png



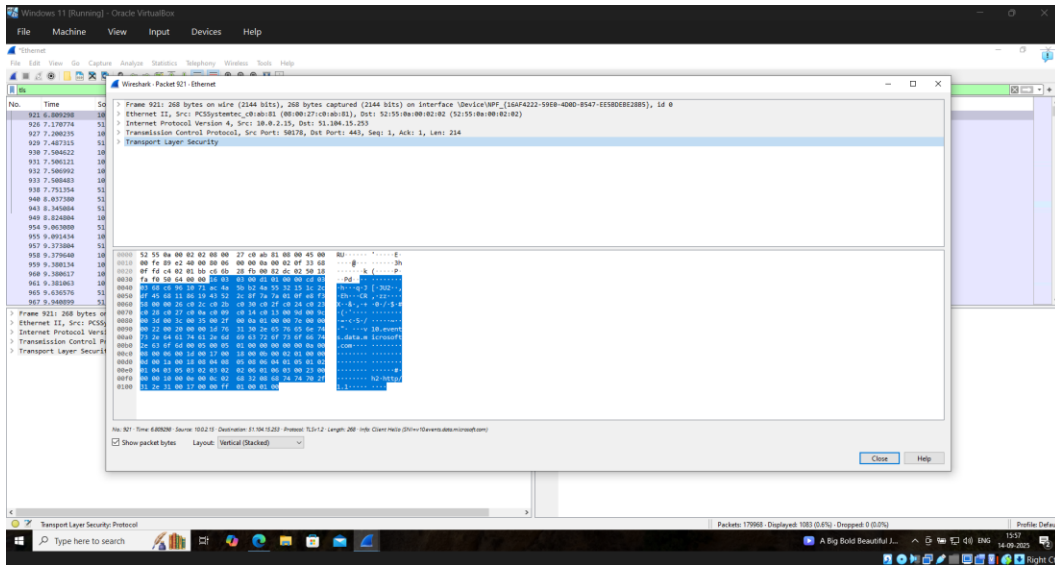
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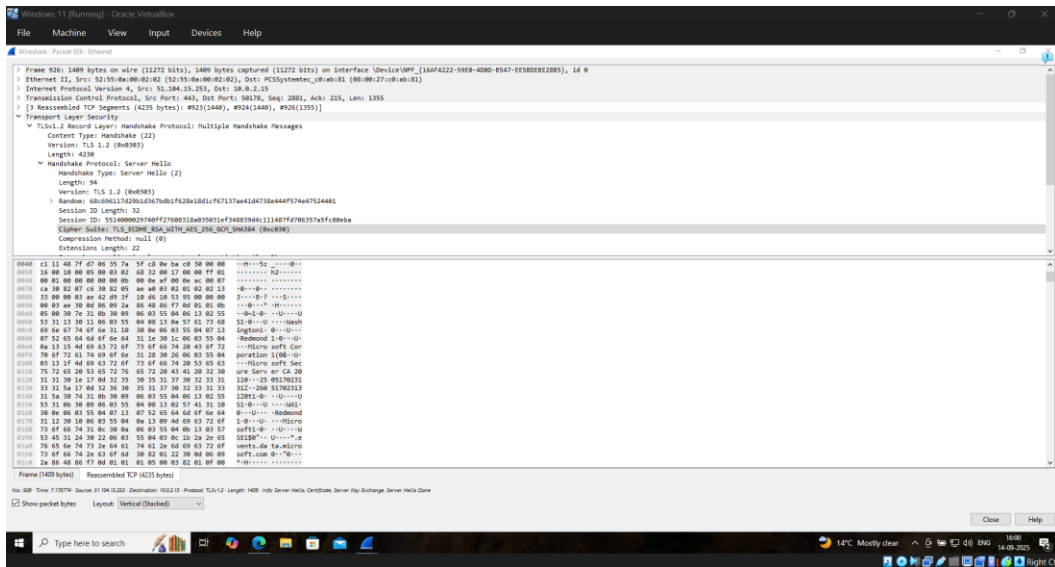
Screenshot 2025-09-14 155536.png



Screenshot 2025-09-14 155721.png



Screenshot 2025-09-14 160059.png



3. Conclusion

Task 1 successfully demonstrated the fundamentals of securing a small network and monitoring traffic. Implemented measures (router hardening, WPA2, disabling WPS, enabling firewall) reduced the risk of unauthorized access. Wireshark captures validated DNS resolution, HTTP requests, and TLS handshakes, confirming that sensitive communications use encryption. For larger networks, additional controls such as IDS/IPS, VPNs, centralized logging (SIEM), and segmentation would be recommended.

4. Closing Remarks

I acknowledge that all work performed relates to my learning objectives and adheres to the organization's confidentiality requirements. I appreciate the opportunity to learn practical network security fundamentals during this internship.